

QUANTUM SEED GENERATOR — Hardware List

All components of the reference build (Raspberry Pi 5). Prices: market level August 2026 (spot-checked live). Total cost excluding monitor/keyboard: approx. €280–310 / ~\$210–300.

#	Component	Details	€	USD
1	Raspberry Pi 5 (4 GB)	Reference platform; 4 GB is fully sufficient; 2026 price level strongly elevated by the RAM price crisis (*USD: \$75 = official reseller list price, market prices often much higher)	€160–175	\$75–160*
2	Official 27 W USB-C power supply	5 V / 5 A for stable Pi 5 operation	€12–15	~\$14
3	microSD card (32 GB, A1, brand quality)	fresh and dedicated to this project (Raspberry Pi OS bookworm); 32 GB is plenty — the system stays small and no data is ever written by the generator; A1 class (e.g. SanDisk Ultra) is the common standard at this size	~€18	~\$20
4	Geiger counter kit RadiationD-v1.1 (CAJOE)	kit incl. J315 tube AND USB power supply (AliExpress); measures background radiation — NO radioactive source included or required	€38	~\$41
5	MLX90640 thermal camera module	AliExpress (I2C, address 0x33; for the GY-MCU90640 variant: PS pin to GND = I2C mode), alternatively Waveshare	~€30–40	~\$32–43
6	Resistor 10 kΩ	1/4 W, metal film; in series in the signal line P3 VIN → GPIO17	< €1	< \$1
7	Jumper wires (female-female)	6 pcs: yellow/violet/green/blue (camera), black/white (CAJOE)	~€5	~\$5
8	Precision dice	1-3 pcs, casino grade recommended (sharp edges, homogeneous material)	~€18	~\$20
9	Monitor without smart functions	plain HDMI monitor, NO smart TV (security rule 3); short, shielded HDMI cable	on hand	on hand
10	USB keyboard (wired)	no wireless keyboard (radio link = interception surface)	on hand	on hand
11	Ethernet cable (setup only)	once for install_seedgen.py; unplugging is enforced before seed generation	on hand	on hand
12	Optional: cooler / enclosure	official Active Cooler; aluminium enclosure with a dumb display possible (check GPIO cable routing)	€7–100	\$8–110

Note: No radioactive source included — only natural background radiation is measured. Wiring: docs/connection_camera.svg, docs/connection_cajoe.svg, or python3 install_seedgen.py --pinout.